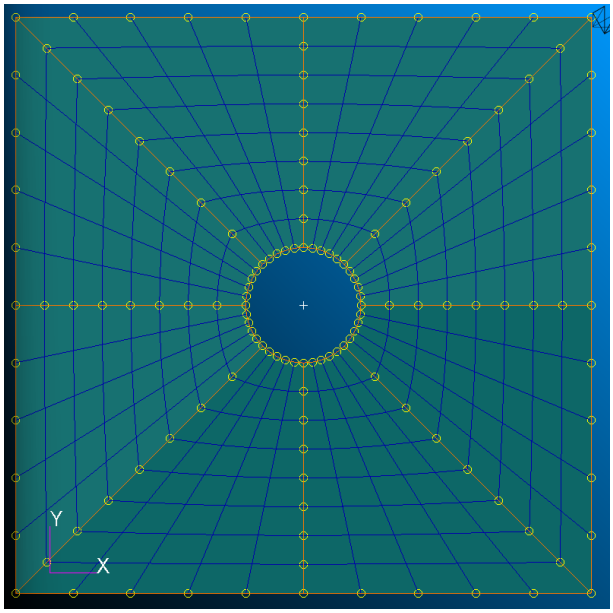
ELS2 - Mé422

CALCUL NUMERIQUE EN MECANIQUE DES STRUCTURES

TP 2 : CALCUL LINEAIRE D'UNE PLAQUE MINCE TROUEE



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1 Model Justification

Consider a square plate on which a tensile force is applied, distributed on each side of the plate :

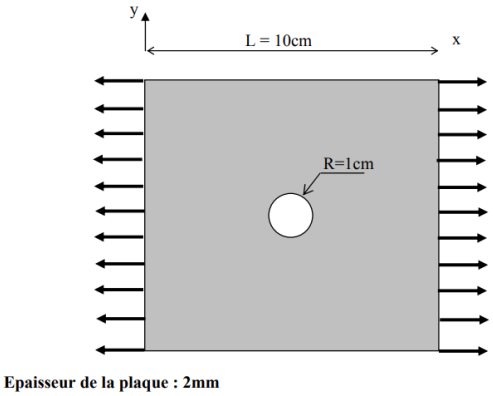


FIGURE 1 – Square plate section

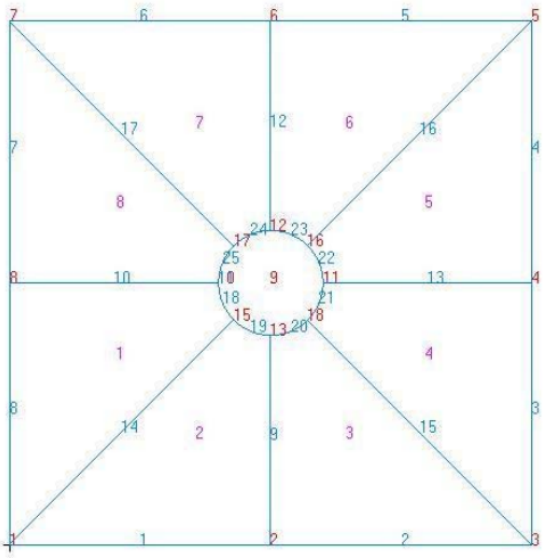


FIGURE 2 – Square plate geometry

Since the most significant deformations are along the x and y-axis, we can reduce the problem to a two-dimensional one to study our case.

Here are the characteristics of the steel we need :

$$\begin{aligned} \rho &= 7,8 \times 10^{-6} \text{kg/mm}^3 \\ E &= 2.1 \times 10^5 \text{ MPa} \\ \nu &= 0.33 \\ \sigma_{R0.2} &= 660 \text{ MPa} \\ \sigma_{RM} &= 849 \text{ MPa} \end{aligned}$$

2 Tensile test

2.1 Analytical calculation :

The objective of this section is to calculate internal forces, deformations, and stresses. The normal stress equation gives us the following :

$$\sigma(x) = \frac{F}{S} \tag{1}$$

Area calculation :

$$S = e * l = 100 \times 2 = 200\,\text{mm}^2$$

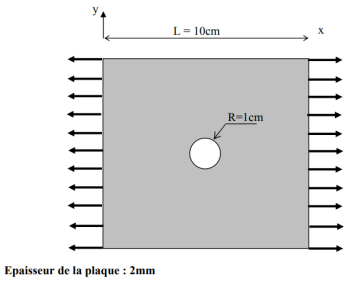


FIGURE 3 – Beam's Section

Then with a force of 10 000N we have :

$$\sigma(x) = \frac{F}{S} = \frac{F}{e * l} = \frac{10000}{100 * 2} = 50Mpa \tag{2}$$

Theoretical calculation of the overstress coefficient around the hole $\frac{a}{w}$:

$$\frac{a}{w} = \frac{2 * R}{L} = 0.2 \tag{3}$$

So with the annex, we can find the coefficient Kf. :

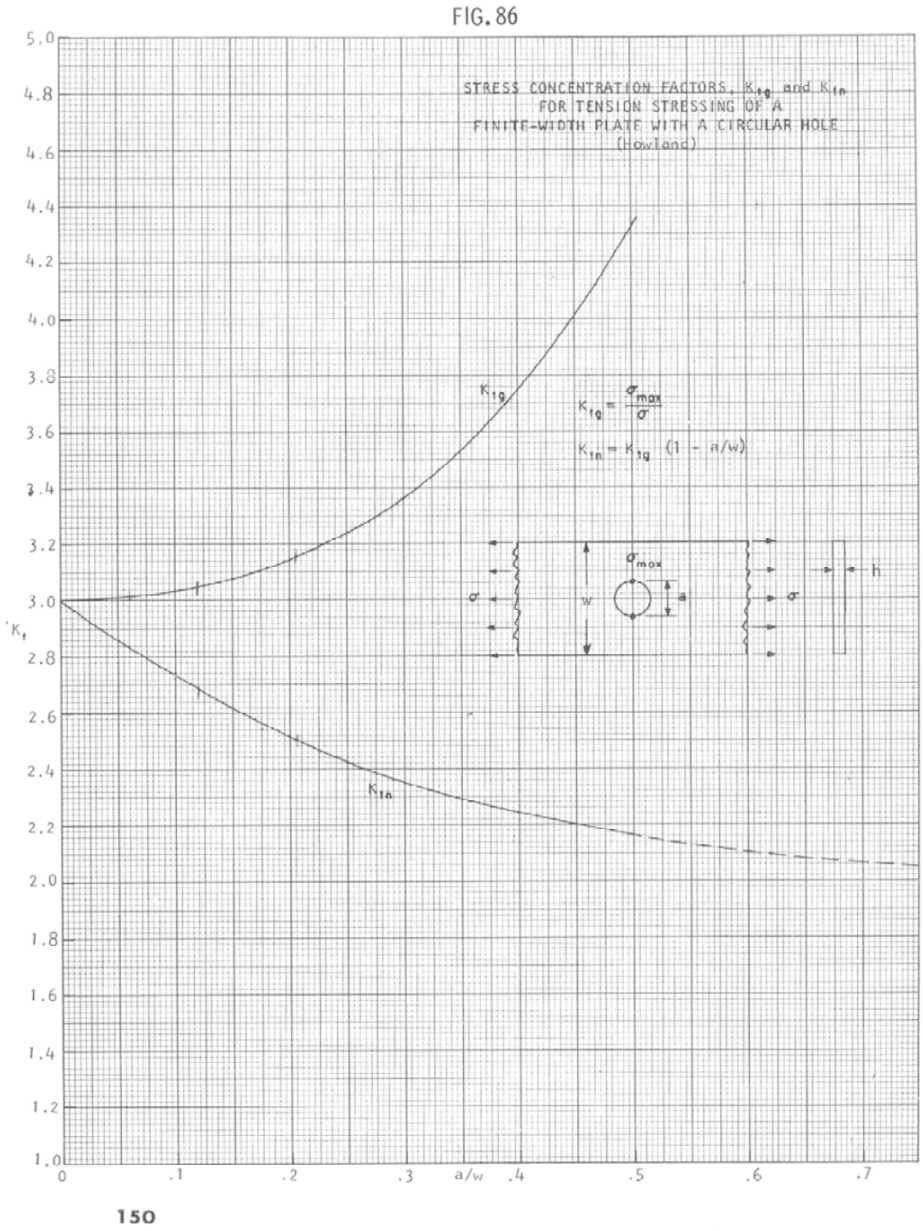


FIGURE 4 – Annex

$$Kf = 3.17 \tag{4}$$

2.2 Analytical results

— $\sigma = \dfrac{F}{S} = \dfrac{10000}{200} = 50\,\text{Mpa}$

— $Kt = 3.17$

— $\sigma_{max} = Kt * \sigma = 3.17 * 50 = 158\,\text{Mpa}$

Interpretation of the results :

The applied stress is :

$$\sigma_{max} = 158\ \text{MPa} < \sigma_{0.2} = 660\ \text{MPa}.$$

Thus, the applied stress is way lower than the conventional yield strength $\sigma_{0.2}$.

The material therefore remains in the elastic domain. No permanent plastic deformation occurs.

2.3 Nastran/Patran modeling

In this section, we detail the construction of the finite element model of the perforated plate.

2.3.1 Geometry modeling

To create the geometry, we first define the square contour of side $L = 100\,\text{mm}$ using points and straight curves.

Then, a circular hole of radius $R = 10\,\text{mm}$ is created at the center of the plate. The circle is modeled using several arc segments in order to allow a structured mesh.

The geometry is divided into eight four-sided surfaces to facilitate meshing.

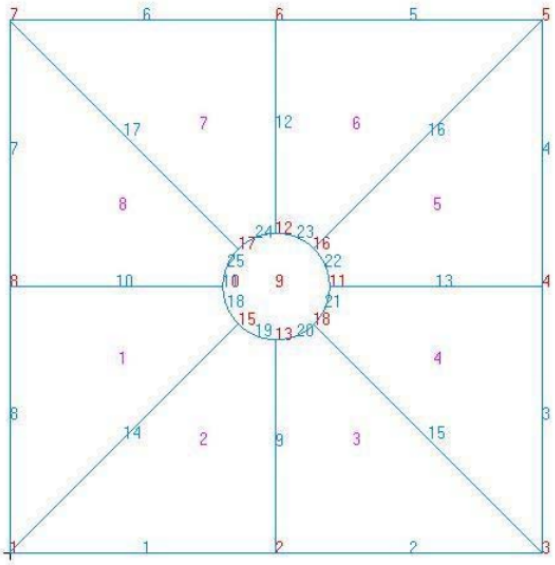


FIGURE 5 – Geometry of square

2.3.2 Mesh modeling

In this part, we define the mesh parameters using the *Mesh Seed* option.

A structured mesh is generated using :

— 2D Quad4 elements
— Isomesh algorithm

A finer mesh is applied around the hole in order to better capture the stress gradient in this critical area.

An **Equivalence** operation is performed to eliminate duplicate nodes and ensure continuity between surfaces.

2.3.3 Boundary Conditions

Now we apply the boundary conditions :

— At $x = L/2$, translations along x and z are blocked
— At $y = L/2$, translations along y and z are blocked

These constraints prevent rigid body motion while allowing the plate to deform under tensile loading.

2.3.4 Loading

A distributed tensile load is applied on opposite sides of the plate.

The total applied force corresponds to 10 000 N distributed along the loaded edges with a force of 500N at the both edge and 9*1000N spread out along the entire length .

2.3.5 Define materials

This part defines the material properties using the following values :

$$\begin{aligned} \rho &= 7.8 \times 10^{-6} \text{kg/mm}^3 \\ E &= 2.1 \times 10^5 \text{ MPa} \\ \nu &= 0.33 \\ e &= 2 \text{ mm} \end{aligned}$$

The material is defined as isotropic steel.

Since the structure is a thin plate, we use **2D shell elements** with the specified thickness.

2.3.6 Results

Here is the result for the displacement field of the plate :

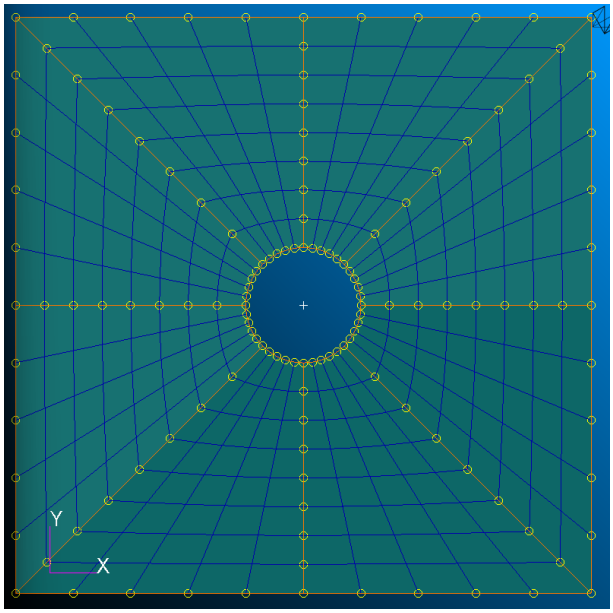


FIGURE 6 – Ending modelisatinon

2.4 Study n°1 – Initial Mesh

2.4.1 Maximum stresses

The maximum Von Mises stress is also concentrated at the boundary of the hole and reaches approximately :

$$\sigma_{VM,max} \approx 146 \text{ MPa}$$

2.4.2 Comparison with theoretical value

From Peterson's stress concentration charts, for :

$$\frac{a}{w} = 0.2$$

the stress concentration factor is :

$$K_t = 3.17$$

The theoretical maximum stress is therefore :

$$\sigma_{max}^{theoretical} = K_t \cdot \sigma_{nominal} = 3.17 \times 50 = 158.5 \text{ MPa}$$

The numerical and analytical values are in good agreement.

2.4.3 Explanation of differences

There is a discrepancy between the numerical value obtained with the initial mesh and the theoretical value of approximately 8.2

This discrepancy can mainly be explained by the mesh density around the hole. The area around the opening has a very high stress gradient. With a mesh that is still relatively coarse, the elements do not allow the stress peak to be captured accurately, leading to a slight underestimation.

In addition, the circular geometry is approximated by quadrangular elements, which introduces a small additional approximation.

Finally, the analytical solution assumes an infinite plate, whereas our model has finite dimensions, which may also have a slight effect on the result.

2.5 Study n°2 – Refined Mesh

In this second study, the mesh density is increased near the hole in order to better capture the stress gradient in the critical region.

2.5.1 Mesh description

The following refinements were applied :

— Increased number of elements along the hole boundary
— Smaller element size near the opening
— Structured quad mesh maintained

The mesh is progressively refined toward the hole to improve stress accuracy without excessively increasing computation time.

2.5.2 Maximum stresses

With the refined mesh, the maximum Von Mises stress slightly increases :

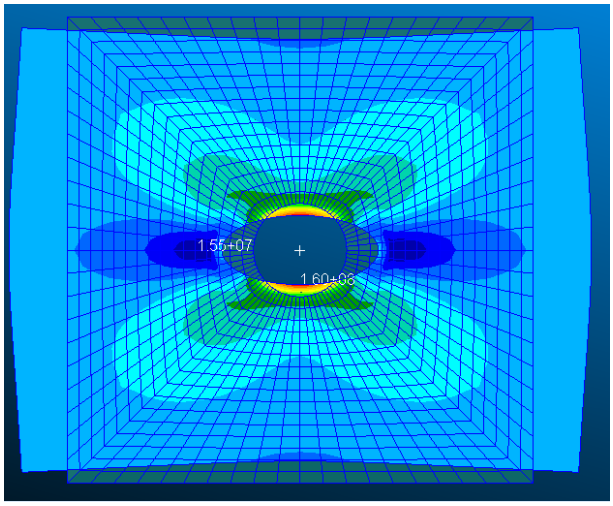


FIGURE 7 – Von Mises stress distribution for the quarter plate model

The maximum value obtained is :

$$\sigma_{max}^{FE,refined} \approx 160 \text{ MPa}$$

2.5.3 Discussion of the evolution

After refining the mesh around the hole, the maximum stress increases slightly.

This behavior is expected in stress concentration problems. A finer mesh better represents the significant gradient present at the edge of the opening. The numerical result then approaches the theoretical value.

This shows that the solution converges as the mesh density increases and confirms the importance of local refinement in critical areas.

2.6 Study n°3 – Quarter plate

In this third study, only a quarter of the plate is modeled by taking advantage of the geometrical and loading symmetries. This allows us to significantly reduce the number of elements and the computation time while keeping the same mechanical behavior.

Symmetry boundary conditions are applied on the two cut planes :
- On the vertical symmetry plane : normal displacement is blocked
- On the horizontal symmetry plane : normal displacement is blocked

2.6.1 Mesh description

A structured Quad4 mesh is used. As in the refined full-plate model, the mesh is highly refined near the hole to correctly capture the strong stress gradient. Far from the hole, a coarser mesh is sufficient since stress variations are much lower.

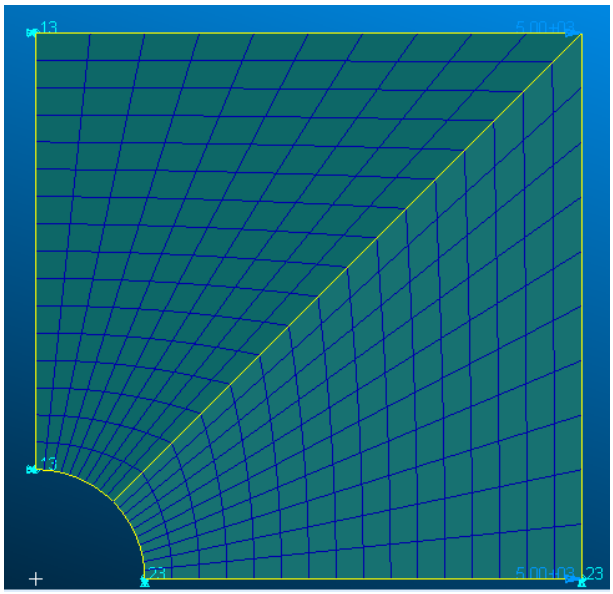


FIGURE 8 – Quarter plane

2.6.2 Maximum stresses

The maximum Von Mises stress is located at the edge of the hole, exactly where the stress concentration is expected.

The maximum value obtained is :

$$\sigma_{max}^{FE,quarter} \approx 160 \text{ MPa}$$

2.6.3 Comparison with the refined full model

The maximum stress obtained with the quarter model is almost identical to the value obtained previously with the refined full-plate mesh.

This shows that :
- The symmetry conditions are correctly applied,
- Modeling only a quarter of the structure does not change the stress level,
- The solution is already mesh-converged.

3 General conclusion

This study analyzed the phenomenon of stress concentration around a hole in a plate subjected to tension.

$$\text{Error } (\%) = \frac{|\sigma_{max}^{FE,refined} - \sigma_{max}^{Theoretical}|}{\sigma_{max}^{FE,refined}} \times 100 = \frac{|160 - 158.5|}{160} \times 100 = 1\%$$

We can see that the numerical and analytical results are very close, this shows a good convergence after mesh refinement.

It is clear that accuracy depends heavily on mesh quality, particularly in areas where stresses vary rapidly The maximum stress remains well below the elastic limit of the material. The plate therefore operates within the elastic range and the design is satisfactory for the applied load.